

SUMMARY

Computer Science B.Sc. student at Tel Aviv University in **final semester (Cumulative GPA: 87.70)**, specializing in **low-level system software, C/C++ hardware simulation engines**, and **automated Python validation harnesses**. Experienced in **POSIX concurrency, Linux kernel debugging**, and **C-Python API integration**.

EDUCATION

B.Sc. Computer Science | Tel Aviv University
Expected Graduation: January 2027 (Final Semester) | Cumulative GPA: 87.70
Selected High-Honors Coursework:

- Software Project / **Multi-threading** & C-Python API (**97**)
 - Introduction to Machine Learning (**100**)
 - Foundations of **Deep Learning** (**96**)
 - Software 1 **Java/OOP** (**97**)
- **Operating Systems** (**95**)
 - Extended Intro to CS - Python (**95**)
 - Algorithms (**95**)
 - Data Structures (**87**)
- **Specialized Coursework:** Computer Structure, Advanced Topics in C++, Systems Design & Networking, Distributed Systems Fundamentals, Optimization for CS.

TECHNICAL SKILLS

- **Systems & Validation:** **C, C++, Linux Kernel Modules (LKM), POSIX Concurrency** (Mutex/Spinlocks), **System Calls, Hardware Mock Layers, Memory Boundary Verification**.
- **Scripting & Automation:** **Python** (Proficient), **C-Python API, High-Throughput Telemetry Log Parsing, Automated Regression Harnesses, Bash Scripting, Linux CLI**.
- **Frameworks & Tools:** **GTest, GMock, Docker, Git/GitHub, SQL Logging, Makefiles, Cursor, Copilot**.

TECHNICAL PROJECTS

- Automated C++ Simulation Engine & Validation Harness (C++20 / Python / GTest)**
- Architected an **Object-Oriented hardware simulation engine** in **C++20** optimized for **spatial cache locality, memory pointer manipulation**, and **mock execution layers**.
 - Implemented **63+ automated unit and integration regression tests** using **GTest and GMock** to systematically isolate **software/hardware edge-case defects**.
 - Built high-throughput **Python automation scripts** to parse, ingest, and analyze **system telemetry logs** and simulation test metrics in a **Linux environment**.
- Linux Kernel Driver Module & POSIX Network Shell (C / Linux CLI / Bash)**
- Engineered custom **Linux Kernel Modules (LKM)** interfacing directly with **system calls** to evaluate **memory boundary safety** and **resource stability** under stress.
 - Developed a **concurrent POSIX socket server** and **UNIX shell** with strict **thread synchronization (mutexes, spinlocks)** to eliminate **deadlocks and race conditions**.
 - Programmed automated **Bash triage hooks** and **stress scripts** to validate **network socket stability** and **kernel resource utilization** during regression runs.
- Distributed Deep Learning Pipeline & Containerized Workflows (Python / PyTorch / Docker)**
- Engineered modular **Python scripts** to parse, clean, and ingest **telemetry data** for **automated pipeline execution**.
 - **Containerized execution environments** using **Docker** to guarantee **100% reproducible computational runs** across multi-system validation setups.

WORK EXPERIENCE & VOLUNTEERING

- Software Engineering & Systems Tutor | Self-Employed**
- Instructed **15+ university students** in **C/C++, Java OOP, Python algorithms, memory management**, and **multithreaded POSIX debugging**.
- Magen David Adom (MADA) | Emergency Medical Responder**
- Managed emergency medical crisis situations for over 2 years in **high-stress, rapid-response team environments**.